

Accuracy Metrics Report

Security Linter — False Positive / False Negative Analysis

Generated	2026-07-23 20:07:12 UTC
Total Cases	99
True Positives	28
True Negatives	37
False Positives	6
False Negatives	28

Comparison of linter output against ground truth defined in samples/ground_truth.json.

	Actually Vulnerable	Actually Safe
Reported VULNERABLE	True Positive: 28	False Positive: 6
Reported SAFE	False Negative: 28	True Negative: 37

Metric	Value
Precision	82.3%
Recall	50.0%
F1 Score	62.2%
False Positive Rate	14.0%
False Negative Rate	50.0%
Accuracy	65.7%

Case	Expected	Actual	TP	FP	FN	Result
C10_dynamic_columns	VULN	VULN	2	0	0	TP
C11_enterprise_app	SAFE	SAFE	0	0	0	TN
C1_simple_assignment	SAFE	SAFE	0	0	0	TN
C2_sql_concatenation	VULN	VULN	1	0	0	TP
C3_sql_fstring	VULN	VULN	1	0	0	TP
C4_sql_printf	VULN	VULN	1	0	0	TP
C5_safe_sanitizer	SAFE	SAFE	0	0	0	TN
C6_augassign_conditional	VULN	VULN	1	0	0	TP
C7_orm_injection	VULN	VULN	0	1	1	TP
C8_safe_orm	SAFE	SAFE	0	0	0	TN
C9_complex_propagation	VULN	VULN	2	0	0	TP
cve_21892	VULN	SAFE	0	0	1	FN
cve_33545	VULN	SAFE	0	0	1	FN
cve_57833	VULN	SAFE	0	0	1	FN
cve_59681	VULN	SAFE	0	0	1	FN
cve_64459	VULN	VULN	0	1	1	TP
cve_67644	VULN	SAFE	0	0	1	FN
rb_seceval_089_0001_vuln	VULN	VULN	1	0	0	TP
rb_seceval_089_0002_vuln	VULN	VULN	1	0	0	TP
rb_seceval_089_001_vuln	VULN	VULN	1	0	0	TP
rb_seceval_089_002_vuln	VULN	VULN	1	0	0	TP
rb_sql_001_safe	SAFE	SAFE	0	0	0	TN
rb_sql_001_vuln	VULN	VULN	1	0	0	TP
rb_sql_002_safe	SAFE	SAFE	0	0	0	TN
rb_sql_002_vuln	VULN	VULN	1	0	0	TP
rb_sql_003_safe	SAFE	SAFE	0	0	0	TN
rb_sql_003_vuln	VULN	SAFE	0	0	1	FN
rb_sql_004_safe	SAFE	SAFE	0	0	0	TN
rb_sql_004_vuln	VULN	VULN	1	0	0	TP
rb_sql_005_safe	SAFE	SAFE	0	0	0	TN

rb_sql_i_005_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_006_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_006_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_007_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_007_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_008_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_008_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_009_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_009_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_010_safe	SAFE	VULN	0	1	0	FP
rb_sql_i_010_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_v2_001_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_001_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_v2_002_safe	SAFE	VULN	0	1	0	FP
rb_sql_i_v2_002_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_003_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_003_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_004_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_004_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_005_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_005_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_006_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_006_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_007_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_007_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_008_safe	SAFE	VULN	0	1	0	FP
rb_sql_i_v2_008_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_009_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_009_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_010_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_010_vuln	VULN	SAFE	0	0	1	FN
rb_sql_i_v2_011_safe	SAFE	SAFE	0	0	0	TN
rb_sql_i_v2_011_vuln	VULN	VULN	1	0	0	TP

rb_sqli_v2_012_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_012_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_013_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_013_vuln	VULN	VULN	1	0	0	TP
rb_sqli_v2_014_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_014_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_015_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_015_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_016_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_016_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_018_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_018_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_019_safe	SAFE	VULN	0	1	0	FP
rb_sqli_v2_019_vuln	VULN	VULN	1	0	0	TP
rb_sqli_v2_020_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_020_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_021_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_021_vuln	VULN	VULN	1	0	0	TP
rb_sqli_v2_022_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_022_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_023_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_023_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_024_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_024_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_025_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_025_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_026_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_026_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_027_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_027_vuln	VULN	VULN	1	0	0	TP
rb_sqli_v2_028_safe	SAFE	SAFE	0	0	0	TN
rb_sqli_v2_028_vuln	VULN	SAFE	0	0	1	FN
rb_sqli_v2_029_safe	SAFE	VULN	0	1	0	FP

rb_sql_i_v2_029_vuln	VULN	VULN	1	0	0	TP
rb_sql_i_v2_030_safe	SAFE	VULN	0	1	0	FP
rb_sql_i_v2_030_vuln	VULN	VULN	1	0	0	TP

case7_orm_injection

[FP] sink=cursor.execute line=20 arg=query — not in expected sink lines [21]

[FN] expected sink=cursor.execute line=21 — not detected

cve_21892

[FN] expected sink=execute line=0 — not detected

cve_33545

[FN] expected sink=execute line=0 — not detected

cve_57833

[FN] expected sink=execute line=0 — not detected

cve_59681

[FN] expected sink=execute line=0 — not detected

cve_64459

[FP] sink=cur.execute line=42 arg=query — not in expected sink lines [0]

[FN] expected sink=execute line=0 — not detected

cve_67644

[FN] expected sink=execute line=0 — not detected

rb_sql_i_003_vuln

[FN] expected sink=execute line=0 — not detected

rb_sql_i_008_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_010_safe

[FP] sink=cur.execute line=11 arg=<expr> — not in expected sink lines []

rb_sqli_v2_002_safe

[FP] sink=db.execute line=23 arg=query — not in expected sink lines []

rb_sqli_v2_002_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_003_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_004_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_005_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_006_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_007_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_008_safe

[FP] sink=cursor.execute line=8 arg=query — not in expected sink lines []

rb_sqli_v2_008_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_009_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_010_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_012_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_014_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_015_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_016_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_018_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_019_safe

[FP] sink=cursor.execute line=14 arg=query — not in expected sink lines []

rb_sqli_v2_020_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_022_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_023_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_024_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_025_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_026_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_028_vuln

[FN] expected sink=execute line=0 — not detected

rb_sqli_v2_029_safe

[FP] sink=cursor.execute line=13 arg=query — not in expected sink lines []

rb_sqli_v2_030_safe

[FP] sink=cursor.execute line=6 arg=query — not in expected sink lines []